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PALEOMAGNETIC EVIDENCE FOR LATE MIOCENE COUNTERCLOCKWISE ROTATION OF NORTH COAST CARBONATE SEQUENCE, PUERTO RICO

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Abstract. A paleomagnetic study of the essentially undeformed middle Tertiary carbonate sequence along the north coast of Puerto Rico reveals statistically significant pre-Pliocene discordance of characteristic component directions against those expected from cratonic North America for much of the section. Despite generally weak to moderately weak magnetic intensities, confirmation of the magnetization as primary in origin comes from the presence of two distinct components of magnetization, intrasite bipolarity, and/or the reproducibility of measurements.

The mean geographic direction for the upper Oligocene to middle Miocene strata is 335.2°/32.9° and the corrected mean paleomagnetic pole is 207.6°/66.5°, (N = 3, α95 = 4.3°). This suggests a counter-clockwise (CCW) block rotation of Puerto Rico and its microplate of 24.5° (± 5.8°) during the late Miocene. Using a width of 250 km for the Northern Caribbean Plate Boundary Zone (NCPBZ) between the North American Plate and Caribbean Plate, the mean left lateral displacement implied is 1.8 to 2.4 cm/yr, which agrees fairly well with published relative motion rates for the two plates. Average rotation rate for 50 Ma to 20 Ma was 0.7°/my but perhaps as great as 4°/my in the Miocene. Resolution of mean paleolatitude indicates northward motion of a degree or less during the period of rotation.

Causes of this short-lived rotation may include (1) tectonic escape from the inhibiting presence of the Bahama Banks and Beata Ridge during eastward motion of Puerto Rico along the sinistral transpressive Puerto Rico Trench and Muertos Trough fault systems or (2) changes in relative plate motions of the Caribbean and North American Plate during the late Miocene.

INTRODUCTION

The island of Puerto Rico, at the eastern edge of the Greater Antilles, is located within the broad, structurally diffuse zone between the North American Plate and Caribbean Plate (NCPBZ; Figure 1). In the present-day configuration of Puerto Rico, the long direction of the island is aligned with the east-west trending Puerto Rico Trench and Muertos Trough, which represent the north and south boundaries of the NCPBZ, respectively. To the west of Puerto Rico, across the Mona Passage, lies the structurally complex island of Hispaniola; to the east lies the Virgin Islands and the northernmost Lesser Antilles across the Anegada Passage. Current models of relative plate motion suggest a tectonic regime dominated by sinistral strike-slip motion with some component of southward convergence on the Puerto Rico Trench, and northward convergence on the Muertos Trough accompanied by sinistral motion [Sykes et al., 1982; Gordon and Jurdy, 1986; McCann et al., 1987].

The structural core of Puerto Rico, which is composed of complexly deformed, Upper Jurassic to upper Eocene volcanic, plutonic, and metasedimentary units, is overlain uncomformably along the north and south coasts by middle Tertiary carbonates and siliciclastic rocks. On the north coast, the deposition of shallow-water carbonates with a minor clastic component was nearly continuous from late Oligocene through Neogene, interrupted by an unconformity during the late Miocene. These units have remained essentially undisturbed structurally, dipping northward four to eight degrees toward the Puerto Rico Trench, where the units become slightly thicker [Monroe, 1980; Seiglie and Moussa, 1984; Moussa et al., 1987].

The sedimentary units on the north coast are; the San Sebastian Formation (late Oligocene), the Lares Formation (late Oligocene to early Miocene), the Montebello Limestone member (early Miocene) and the coeval to slightly younger Cibao Formation, the Los Puertos Limestone (middle Miocene), the Aymamon Limestone (middle to early late Miocene), and the Quebradillas Limestone (early Pliocene) (Figure 2). Variations in the composition and provenance of these units can be approximately related to the changing paleo-water-depths caused by global transgressive-regressive cycles superimposed on a uniform 2-kilometer subsidence of the north coast during the Oligocene and Miocene epochs [Seiglie and Moussa, 1984; Birch, 1986].

PURPOSE OF THIS STUDY

Previous paleomagnetic studies on Puerto Rico by Fink and Harrison [1972], Elston and Krushensky [1983], and Van Fossen et al. [1989] have been confined to the Cretaceous to Eocene core of the island, with CCW rotations measured at 58° post-Cretaceous, 83° post-Eocene, and 45° post-Eocene respectively. These studies were conducted in structurally complex areas. To prevent the structural factor from biasing the paleomagnetic results, we sampled the undeformed carbonate units along the north coast to determine the amount and timing of rotational tectonics affecting Puerto Rico.

In the summer of 1987, 268 cores at 38 sites were collected from five of the six carbonate units along roadcuts on the northwest coast of Puerto Rico (Figure 2). Excellent

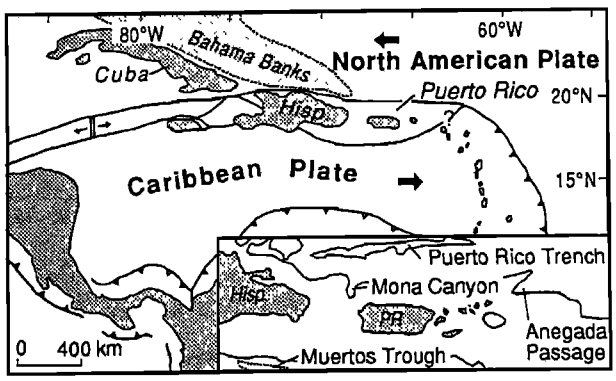


Fig 1. Tectonic map of the Caribbean plate area.

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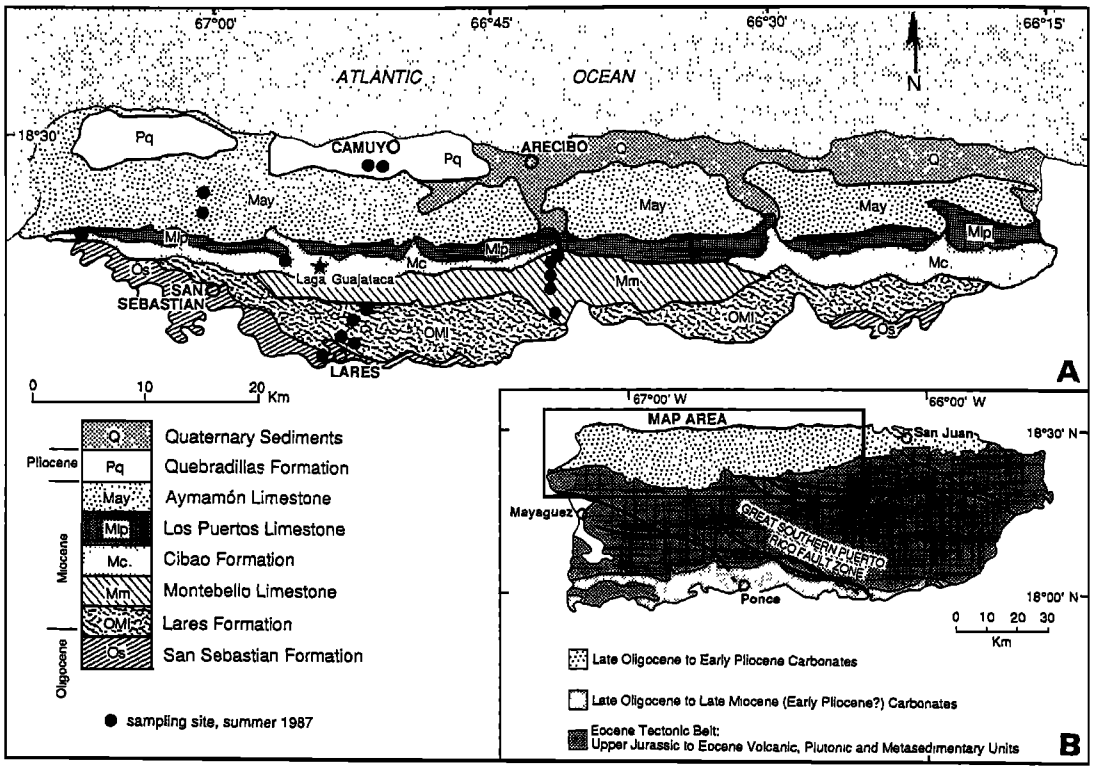


Fig 2. A) Geologic map of the north coast, Puerto Rico. Paleomagnetic sampling sites denoted as solid dots. B) Insert - geologic map of the island of Puerto Rico.

exposure allowed sampling in the entire section of the Lares Formation and the Montebello Limestone Member. However, several sites were unusable due to low magnetic mineral content (NRM's < $8x10^{-8}$ emu/cc) or complications due to secondary weathering products. The middle and upper sections of the Lares Formation, the middle and upper sections of the Montebello Limestone Member, and the Cibao Formation proved least useful in this regard. In most of the other sites in these units, only one or two cores yielded reliable results. Table 1 contains site means and statistics.

All cores were drilled with a hand-held, gas-powered, water-cooled drill and were oriented in the field, with a minimum of six cores per site. All cores were stored immediately upon arrival in the 'field-free' room at Syracuse University, where the mean ambient field strength is 200 gammas. Demagnetization of cores was done within the 'field-free' room using a SSM-2 Schonstedt Spinner Magnetometer, Schonstedt thermal demagnetizer, and Sapphire Instruments SI-4 Alternating Fields demagnetizer.

Isothermal remanent magnetization (IRM) acquisition experiments and demagnetization of saturation remanence suggest that low coercivity multi-domain titanomagnetite is the primary carrier of NRM. The cores display overlapping temperature spectra with low to moderate blocking temperatures for the secondary component (150°-350°) and moderate to high blocking temperatures for the characteristic component (220°-550°C). In some sites, the primacy of magnetization for the characteristic components is confirmed by the presence of both normally and reversely magnetized directions. These populations are antiparallel at the 95% confidence level.

SITE MEAN DIRECTIONS, VGPs, and ROTATION

Because only a few cores were available for characteristic component analysis in 8 of the 22 sites, conventional procedures were deemed unacceptable. Two of the nine sites

TABLE 1: Mean Geographic Directions - Characteristic Component.

Site	n/N		Dec	Inc	α95	k
San Sebastian Formation						
87R001	5/6		320.5	27.0	20.7	14.6
87R002	5/8		339.9	40.8	22.4	12.6
Lares Formation						
87R003	3/4 ◊		333.6	39.4	32.8	15.2
87R004	2/2 ◊		341.7	34.1	-	-
87R005	5/6		347.0	32.2	23.5	11.6
87R005	1/1 ◊		337.9	2.3	-	-
Montebello Limestone						
87R022*	3/3 ◊		168.7	28.1	97.1	2.7
87R021	3/3 ◊		337.8	19.7	26.9	22.1
87R024	2/2 ◊	B	150.2	-18.8	-	-
87R023*	2/2 ◊		89.6	36.5	-	-
87R027*	1/1 ◊		205.0	25.2	-	-
87R028	2/2 ◊		327.8	7.0	-	-
87R029	1/1 ◊	B	151.0	-46.8	-	-
87R033	1/2 ◊		309.7	29.2	-	-
87R034	2/2 ◊		334.7	47.3	-	-
Cibao Formation						
87R026	4/4 ◊		335.7	46.5	24.9	14.6
Aymamon Limestone						
87R037	3/3 ◊		338.2	19.1	34.2	14.0
87R040	5/5 ◊	B	157.2	-2.8	10.4	55.3
Quebradillas Limestone						
87R045	6/7	B	185.4	-44.0	20.6	11.6
87R044	5/6	B	177.5	-33.9	25.7	9.8
87R047	5/7	B	180.1	-22.9	27.6	8.7
87R046	6/6	B	171.7	-9.4	30.5	5.8

* = Data not used in later analyses.
B = Most or all directions are reversed polarity.
n/N = Stable characteristic components/Samples analysed
◊ = Sites in which NRM's of additional cores were too weak for demagnetization analysis.

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of Montebello Limestone contained three samples with stable remanence. The other seven sites had less than three samples with a stable remanence (Table 1). To prevent bias, data from all cores with reasonable results have been combined into four time-slots: late Oligocene, which comprises results from the San Sebastian and Lares formations; early Miocene, represented by Montebello Limestone Member and Cibao Formation; middle Miocene for the Aymamon Limestone; and early Pliocene for the Quebradillas Limestone (Table 2). In each time-slot, the cores used were taken from the same location; sites are no more than 0.5 km from each other, and in most cases stratigraphically adjacent (Figure 2). The single Cibao Formation site (87R025) used in the averaging, is the only exception to this.

From figure 3, the three earliest mean directions (upper Oligocene to middle Miocene units) have pervasive northwesterly directions, indicating a CCW rotation relative to North America (expected declinations computed from Irving and Irving, 1982). For the early Pliocene data, only 1.7° of declination difference exists, which is not statistically significant from that expected (Table 2). Differences in declination with associated error limits have been calculated using Coe et al. [1985] (Table 2). Paleolatitudes calculated from the mean directions (with the notable exception of the middle Miocene data set) suggest a paleolatitude not significantly different from present day (Table 2).

The middle Miocene characteristic component data set, which is dominantly reversely magnetized, shows an anomalously shallow inclination (8.7°). This is likely due to either insufficient averaging of secular variation or inclination shallowing during compaction and karstification. It has been noted that an apparent difference exists in inclination between normal and reversed polarities from the Miocene epoch [Diehl et al.,1988], giving an apparent shallowing of reversely magnetized inclinations. A test of statistically significant different directions between the normal and reversed directions (Mean inc=19.0°, n=2; mean inc= -5.5, n=6, respectively), when the latter is inverted through the origin, indicates only a 39 percent probability that both share the same direction. Preserving the distribution of middle Miocene directions and their mean declination, the mean inclination has been rotated to agree with the expected inclination for a paleolatitude of 18°N. The recomputed VGP (205.8°/ 68.7°) is statistically identical to the 27 Ma and 22 Ma VGP's at the 97% and 70% confidence level respectively. The agreement between computed mean declinations for the three data sets suggest that, with respect to the geographic reference frame, no tectonic rotation of Puerto Rico occurred during this time period. Consequently, the three VGP's are averaged to yield a 27 to 11 Ma mean reference pole for Puerto Rico (207.6E, 66.5N, a95= 4.3, K=830.1, paleolatitude=17.8N). The corresponding mean direction is 335.0°/32.7°.

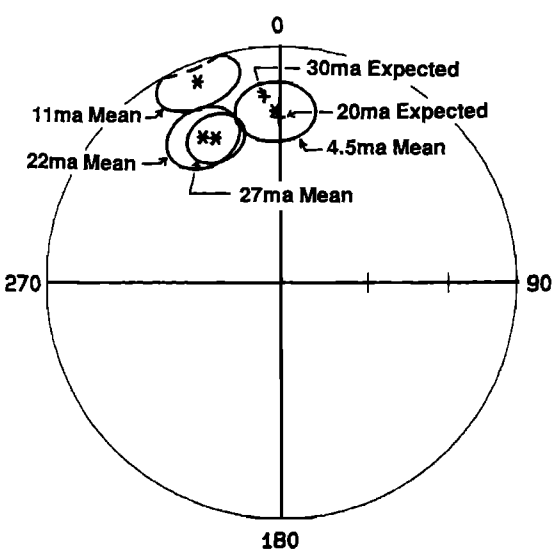


Fig 3. Equal area plot directions for combined samples. Expected reference directions shown as + symbols. Note counterclockwise discordance of 27-11 my data base.

CONCLUSION AND TECTONIC INTERPRETATIONS

Using the mean observed declination direction of 335° for Puerto Rico and the mean expected declination direction for cratonic North America of 359° during the late Oligocene to middle Miocene [Irving and Irving, 1982], we conclude that the island of Puerto Rico tectonically rotated CCW 24° ± 5.8° with respect to North America between approximately 11 and 4.5 Ma. Possible latitudinal translation during the period is statistically undetectable. Figure 4 shows the Eocene pole from Van Fossen et al. [1989], and from this current study, the late Oligocene-early Miocene and the Pliocene reference poles for Puerto Rico, with implied rotation angles.

The paleomagnetic results of this study of the carbonate sequence from the north coast of Puerto Rico suggest that a CCW block rotation, consistent with the bounding sinistral transcurrent fault system [Beck, 1989], occurred within a short amount of time: about 6 to 7 million years during the late Miocene. Unfortunately, this time slice is coeval with a global sea level low and subaerial erosion of Puerto Rico [Seiglie and Moussa, 1984]; hence this rotation cannot be tracked. The implied rotation rate during this time, 4°/my, is identical to what would be predicted for Puerto Rico if the crustal block behaved as a cylindrical-shaped terrane bounded by strike-slip fault zones. In such zones, with simple shear and 100% coupling, the angular velocity (AV) in radians/ma is given by

AV = Slip Rate (km/ma) / width of shear zone (km)

Using a width of the NCPBZ of 250 km, and reported relative motion rates of 1.3 to 2.4 cm/yr [Sykes et al., 1982; Gordon and Jurdy, 1986] we conclude that the Puerto Rico microplate behaved as a perfect roller-bearing between the North American and Caribbean plates during the period 11 to 4.5 ma.

There is much structural and stratigraphic evidence of tectonic disruption within the NCPBZ that has not been discussed in detail here. However, the results of this study suggest that prior to the late Miocene, the Puerto Rico microplate [Byrne et al., 1985] was aligned with Hispaniola, with the long axis of Puerto Rico trending approximately

TABLE 2. Mean Geographic Directions and VGPs of Characteristic Components for Four Time-Slots.

EPOCH	Age	n/N	Dec	Inc	α95	k	ϕ	λ	α95	Paleoλ	Rotation Δdec
late Oligocene	27ma	21/27	335.9	33.2	9.2	13	211.0	67.2	7.6	18.1	21.4 ±7.9
early Miocene	22ma	15/30	331.9	30.7	11.5	12	206.1	63.7	8.8	16.5	27.8 ±9.7
middle Miocene	11ma	8/8	337.6	8.7	11.5	24	174.0	64.1	7.2	4.4	22.4 ±9.3
corrected m. Miocene			337.6	33.0			205.8	68.7			
early Pliocene	4.5ma	22/27	358.3	28.0	12.0	8	164.2	87.4	10.8	14.9	1.7 ±9.9

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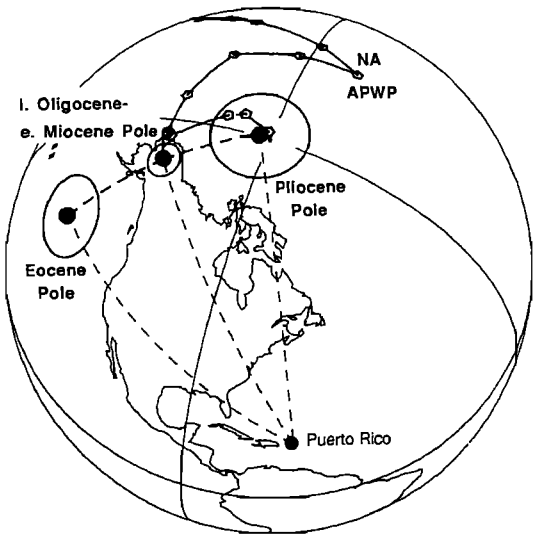


Fig 4. Oblique projection showing paleomagnetic pole for north coast carbonates (age range 27-11 Ma), Eocene pole from southwest Puerto Rico [Van Fossen et al. 1989], and North American Apparent Polar Wander Path. APWP pentagons represent mean poles at 20 ma intervals.

N66°W, similar to that of the Bahama Banks. If the microplate was traveling eastward with the Caribbean Plate proper, it may have been as much as 200 km to the west at the beginning of late Miocene, near the present-day position of eastern Hispaniola. Rotation may have occurred through sinistral motion as the Puerto Rico microplate became uncoupled from Hispaniola and released from the inhibiting effects of the Bahama Banks, Beata Ridge, and the Hispaniola Restraining Bend [Mann et al., 1984]. Alternatively, changes in relative plate motions between the North American Plate and Caribbean Plate at about 9 Ma may have been the catalyst for rotational tectonism [McCann et al., 1987]. Cessation of rotation may have been due to the detachment of the Puerto Rico microplate from the Caribbean Plate proper and transfer of extensional stress to the Anegada Basin and Mona Canyon [Maley et al., 1974; Lithgow, et al., 1987; McCann et al., 1987].

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